

Construction and Characterization of an Analog Lock-In Amplifier

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Lock-in amplification is a widely used technique for measuring weak oscillatory signals in a noisy background. We build an analog single-phase lock-in amplifier on a breadboard using operational amplifiers and RC filtering. The system is characterized by detecting light from a red LED using a photodiode both in the presence and absence of ambient room lighting. The LED is driven sinusoidally by a function generator, which also provides the square-wave reference signal for lock-in detection.

I. INTRODUCTION

We briefly outline the operating principle of analog lock-in amplification, tailored to this project rather than treat it in full generality; for a more complete discussion, see the *Zurich Instruments* white paper [1]. We wish to measure a weak input signal,

$$V_s(t) = A_s \cos(\omega_s t + \phi), \quad \omega_s = 2\pi f_s, \quad (1)$$

and determine its amplitude A_s . In general, the usually weak signal may be buried in noise, preventing direct measurement [2]. The key idea of lock-in amplification is to multiply the signal by a reference and clean it. The reference is here chosen as a square wave,

$$V_r(t) = A_r \operatorname{sgn}[\cos(\omega_r t)]. \quad (2)$$

Because the square reference contains all odd harmonics, the mixed signal contains components at frequencies $\omega_s \pm (2k+1)\omega_r$, for $k \in \mathbb{N}$. After low-pass filtering, only the component at the fundamental detuning $\delta = \omega_s - \omega_r$ remains, producing the *in-phase* output,

$$X(t) = \frac{2\Gamma}{\pi} \cos(\delta t + \phi) \xrightarrow{\delta=0} X = \frac{2\Gamma}{\pi} \cos(\phi), \quad (3)$$

with $\Gamma = A_s A_r$. When "on lock" (i.e. $\delta = 0$), it reduces to a DC signal, from which information about $V_s(t)$ can be inferred. This procedure is known as *demodulation*. For a full derivation of Eq. 3, see the supplementary information.

II. SETUP

A detailed schematic of the setup is shown in Fig. II, while Fig. III shows its breadboard implementation. For analysis, it is convenient to divide the circuit into the five stages indicated by the red icons in Fig. 1. First, an OPT101 photodiode converts light from a nearby sinusoidally driven LED into a current. Its integrated transimpedance amplifier then converts this current into an oscillating voltage, which serves as the input signal $V_s(t)$.

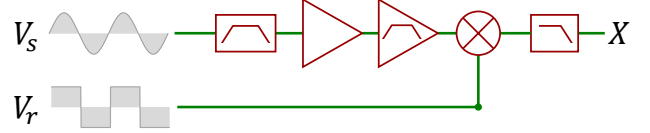


FIG. 1. Simplified diagram of the five stages of the lock-in amplifier chain.

Stage 1: The signal passes through a passive band-pass filter (BPF), formed by cascading a high-pass section (R_7, C_4) with a low-pass section (R_6, C_5), suppressing frequency components far from f_s . The component values give a passband $f \in [1.59, 15.9]$ kHz; see Table I.

Stage 2: The signal is then fed into a non-inverting amplifier built around op-amp IC2A. Assuming a high input impedance, this stage does not significantly load the preceding passive filter. Its closed-loop gain is $G = 101$; see Table I.

Stage 3: Next, the signal enters an inverting active band-pass amplifier. This stage has the effect of an inverting amplifier, but with frequency-dependent input and feedback impedances, so that it both amplifies and further filters the signal. Its passband coincides with that of stage 1, and the mid-band gain is $G = -10$; see Table I.

Stage 4: The reference signal at BNC2 controls two ADG413N analog switches A and B alternately. A high reference closes A, routing the signal through a non-inverting amplifier with *sum* closed-loop gain $G_A = +1$, while a low reference closes B, routing it through an inverting amplifier with closed-loop gain $G_B = -1$. Stage 4 thus multiplies the signal by ± 1 in synchrony with the reference, implementing mixing with $V_r(t)$. This switching-implementation of a mixer is one reason to opt for a square-wave reference.

Stage 5: Finally, the mixed signal is passed through a passive low-pass filter (PLPF). Its corner frequency $2\pi f_c = 1/R_{10}C_7$ sets the detection bandwidth of the system.

Despite what is indicated in Fig. II, the breadboard implementation uses *two* TL074 ICs to reduce wiring complexity: one for stages 2 and 3, and one for stage 4.

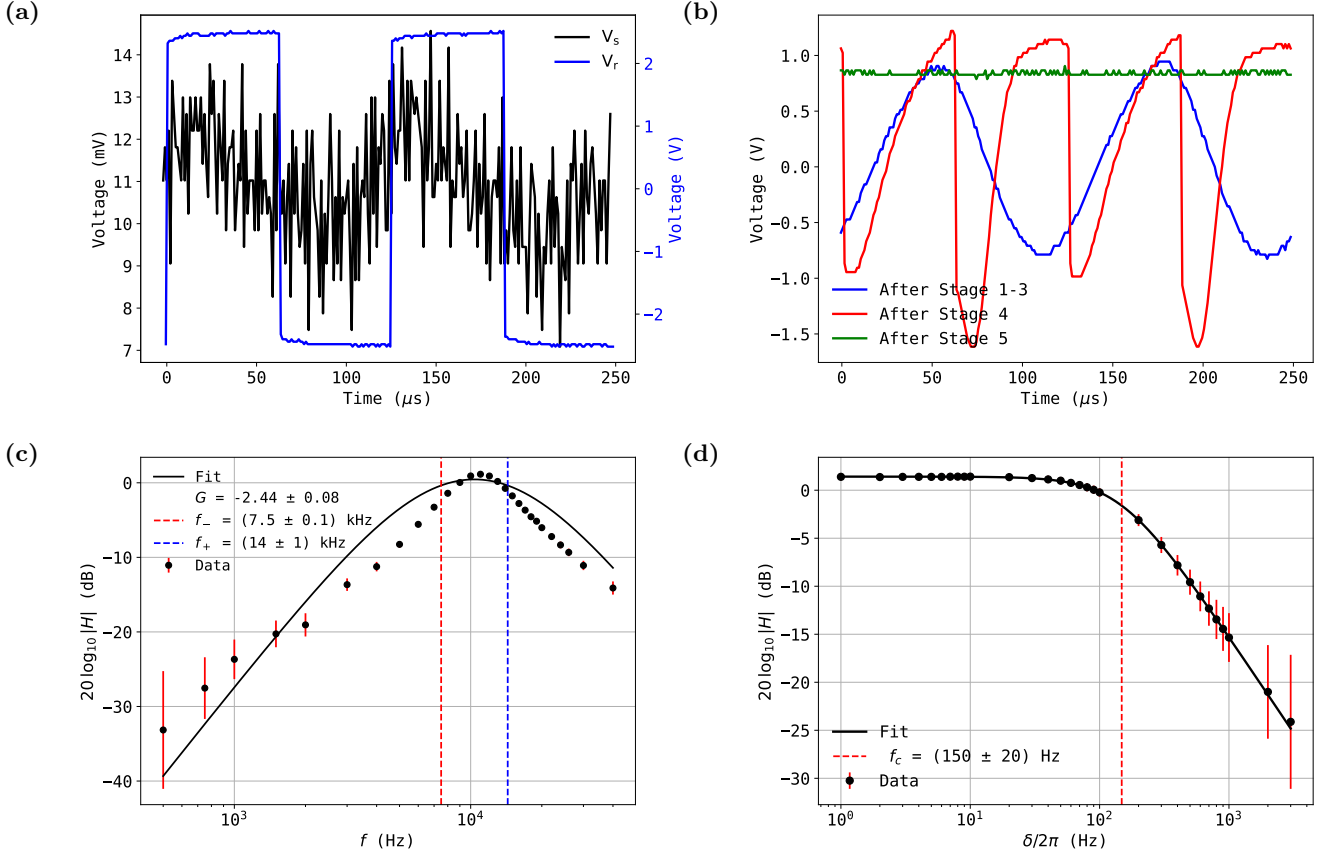


FIG. 2. (a) Input signal V_s (black, left y-axis) from photodiode OPT101 stimulated by the red LED in absence of ambient room light, with square reference signal V_r (blue, right y-axis) supplied by a function generator, at same frequency ($\delta = 0$). (b) Output signal after pre-filtering and amplification, stages 1-3 (blue), after mixing, stage 4 (red) and final lowpass filtering, stage 5 (green). (c) Bode magnitude plot of measurements from experiment 1 along with the fitted theoretical transfer function of stages 1-3. The data is the raw measured output. (d) Bode magnitude plot of measurements from experiment 2 along with the fitted theoretical transfer function of stage 5. The data is the raw measured output.

III. EXPERIMENT & ANALYSIS

Fig. 2(a-b) depicts the time-domain response of the various stages of our setup, as a demonstration of the lock-in amplifier's central function. We use a red LED, driven by the function generator, to stimulate the photodiode, and shield the system using a cardboard box to remove ambient room lighting. After setting the detuning to zero ($\delta = 0$), we measure the output voltage of the photodiode (black, left y-axis) and the reference signal (blue, right y-axis) in Fig. 2(a), and the signals after pre-filtering and amplification (blue, stages 1-3), mixing, stage 4) and post-filtering (green, stage 5) in Fig. 2(b). Upon comparing the figures, we see how stages 1-3 transforms a small noisy millivolt signal to a stronger signal on order volts with higher resolution. Looking at the output of stage 4, we see abrupt changes in the sign of the signal concurrent with the reference signal, a direct result of square-wave mixing. Stage 5 then transforms

the signal to a clear DC-signal, completing the lock-in amplification.

To characterize the lock-in amplifier, we perform two experiments. In both, two function generator channels are used: one drives the LED at frequency f_s , while the other provides the reference at f_r . The output voltage is measured at BNC1 in Fig. II and recorded with a PicoScope[®] 2204A oscilloscope using the PicoScope[®] 7 laptop application.

Experiment 1: The signal and reference frequencies are set equal, $f_s = f_r \equiv f$, and f is swept from 500 Hz to 40 kHz. At each frequency, the reference phase offset θ is adjusted to maximize the DC output, corresponding to the substitution $\phi \rightarrow \phi - \theta$ in Eq. 1. The maximum output occurs at phase-lock, $\phi - \theta_{\max} = 0$. The reference is then shifted by 180° to measure the minimum output, allowing us to correct for any DC offset:

$$X_{\text{offset}} = \frac{X_{\text{max}} + X_{\text{min}}}{2}, \quad X_{\text{amp}} = X_{\text{max}} - X_{\text{offset}}. \quad (4)$$

Since $f_s = f_r$, the mixer output contains a DC component ($\delta = 0$), so stage 5 does not add frequency-dependent attenuation. The measured sweep therefore reflects the combined transfer function of stages 1–3. We plot X_{amp} as a function of f and fit the data to the magnitude of Eq. 6. This yields the cumulative gain and the lower and upper corner frequencies of the band-pass response. The resulting Bode plot is shown in Fig. 2(c), and the extracted parameters are summarized in Table II.

Experiment 2: The signal frequency is fixed at $f_s = 11$ kHz, while the reference frequency f_r is varied such that the detuning $\delta/2\pi = |f_s - f_r|$ is swept from 0 Hz to 3 kHz. At each detuning, the reference phase offset is adjusted to maximize the measured output amplitude, as in experiment 1. Since f_s lies within the passband of stages 1–3, the δ -dependence is set by stage 5 alone. We therefore record the root-mean-square output which is multiplied by $\sqrt{2}$ and fit to

$$X(\delta) = A |H_5(i\delta)|, \quad (5)$$

where A is a constant prefactor absorbing the gain of previous stages. A fit to Eq. 5 yields the low-pass corner frequency f_c . The resulting Bode plot is shown in Fig. 2(d), and the extracted value is listed in Table II.

For both experiments, voltage uncertainties are estimated from the standard deviation of fit residuals. For experiment 1, a constant is fitted to the stage 5 signal and the residual standard deviation is used as the voltage uncertainty in Fig. 2(c). For experiment 2, no waveform was recorded at stage 5, so instead a sine wave is fitted to the stage 3 signal and its residual standard deviation is used as the voltage uncertainty in Fig. 2(d). In both cases, error propagation is applied to convert the voltage uncertainties into decibels.

IV. DISCUSSION

The characterization of stages 1–3 should be interpreted with some caution. In experiment 1, the measured sweep reflects the *combined frequency response of the LED, the photodiode, and stages 1–3*. Our analysis therefore assumes that the LED and photodiode response is approximately flat over the measured frequency range, such that the fit still provides a reasonable estimate of the response of stages 1–3. For stage 5, the contributions of the LED and photodiode is absorbed by A .

For stages 1–3, the lower and upper corner frequencies are too close to produce a clear flat mid-band region. As seen in Fig. 2(c) the fit is also very poor. The first corner frequency f_- deviates significantly from the predicted value, while the second corner frequency f_+ agrees within 2σ . The measured overall gain is also considerably lower than the predicted mid-band value. This discrepancy is attributed to saturation of the amplifier. With a

reference amplitude of $A_r = 5$ V, the amplified signal exceeds the linear operating range of the op-amp, clipping the output and reducing the apparent gain. Redoing the experiment using a different reference amplitude to avoid saturation seems an obvious next step to fully characterize the device. All predicted values are given in Table I.

Possible error sources include the DC offset measured in experiment 1. We were not able to get rid of this. Connecting the ground of the breadboard to a metal board could have removed or diminished it.

We also observe charge spikes in the measured in-phase output $X(t)$. These occurred at both the rising and falling edges of the reference signal. We introduced additional bypass capacitors to all amplifiers and switches, but saw no improvements, see Fig. III.

A natural extension of the present setup would be *dual-phase demodulation*. By measuring both quadrature components, $X(t)$ and $Y(t)$, one could reconstruct the full signal phasor rather than a single projection of it. In the present breadboard implementation, this would require an analog 90° phase-shifting circuit for the reference, together with a second copy of stages 4 and 5 from Fig. II.

V. CONCLUSION

We have constructed an analog lock-in amplifier capable of measuring small optical signals using a photodiode. The amplifier produces a DC output when the signal and reference frequencies are matched, and an AC output at the difference frequency otherwise. Additionally we have characterised the bandpass and low-pass filters of the instrument.

VI. ACKNOWLEDGEMENTS

We thank JB and Jörg for delivering an interesting course and for their help throughout the project! A special thanks to Jörg for providing the schematic shown in Fig. II.

VII. BIBLIOGRAPHY

- [1] Zurich Instruments. Principles of lock-in detection and the state of the art. Technical report, Zurich Instruments, 2023.
- [2] Kenneth Libbrecht, E. Black, and C. Hirata. A basic lock-in amplifier experiment for the undergraduate laboratory. *American Journal of Physics*, 71, 2003.

SUPPLEMENTARY INFORMATION

All data and analysis code associated with this project are available in the [project GitHub repository](#).

Stage	Function	Defining network	G	ω_- (rad/s)	f_- (kHz)	ω_+ (rad/s)	f_+ (kHz)
1	Passive BPF	R_7C_4 (HPF), R_6C_5 (LPF)	–	10^4	1.59	10^5	15.9
2	Non-inv. amp.	R_1, R_2	$1 + \frac{R_2}{R_1} = 101$	–	–	–	–
3	Active BPF	$Z_{in} = R_5 + \frac{1}{sC_3}, \quad Z_f = R_3 \parallel \frac{1}{sC_2}$	$-\frac{R_3}{R_5} = -10$	10^4	1.59	10^5	15.9
5	Passive LPF	$R_{10}C_7$	–	–	–	10^3	0.159

TABLE I. Predicted characteristics of stages 1–3 and 5 in the amplifier chain, based on the component values in Fig. II. Here, G denotes the flat closed-loop gain for non-band-pass stages, and the mid-band gain for band-pass stages. The quantities $\omega_{\pm}$ and $f_{\pm}$ denote the lower and upper corner frequencies, predicted from the relevant RC time constants as $\omega_c = 1/RC$ and $f_c = 1/(2\pi RC)$, where applicable.

Stages	Quantity measured	Prefactor	f_- (kHz)	f_+ (kHz)	Offset (dB)
1–3	BPF + amplification	$ G = -2.44 \pm 0.08$	7.5 ± 0.1	14 ± 1	–
5	LPF response	$A = 1.0 \pm 0.1$	–	0.15 ± 0.02	1.4 ± 0.1

TABLE II. Measured transfer characteristics extracted from the two experiments.

The combined Laplace transfer function of stages 1–3 takes the form of a fourth-order band-pass response with mid-band gain $G = -1010$:

$$H_{1-3}(s) = H_1(s)H_2(s)H_3(s) = \frac{G \cdot (s/\omega_-)^2}{(1 + s/\omega_-)^2 (1 + s/\omega_+)^2}. \quad (6)$$

The ideal transfer function of stage 5 takes the standard form of a first-order passive low-pass filter:

$$H_5(s) = \frac{1}{1 + s/\omega_c}. \quad (7)$$

Derivation of $X(t)$

The input signal from the photodiode is given as:

$$V_s(t) = A_s \cos(\omega_s t + \phi), \quad \omega_s = 2\pi f_s \quad (8)$$

and the reference signals is a square-wave:

$$V_r(t) = A_r \text{sgn}[\cos(\omega_r t)], \quad \omega_r = 2\pi f_r \quad (9)$$

What mixing does is effectively compute $X(t) = V_s(t)V_r(t)$.

Fourier expansion of $V_r(t)$

To analytically determine $X(t)$, we write the Fourier expansion of $V_r(t)$. $V_r(t)$ is a product of two even functions, so its expansion is given by:

$$V_r(t) = A_r(a_0/2 + \sum_{n=1}^{\infty} a_n \cos(n\pi t/L)), \quad n \in \mathbb{N} \quad \& \quad L = T/2 \quad (10)$$

with $T = 2\pi/\omega_r$ being the period of $V_r(t)$, with:

$$a_n = 2/L \int_0^L dt V_r(t) \cos(n\pi t/L) \quad (11)$$

For a_0 , we get:

$$a_0 = 2/L \int_0^L dt V_r(t) \cos(0) \quad (12)$$

$$= 4/T \int_0^{T/2} dt \operatorname{sgn}[\cos(\omega_r t)] \quad (13)$$

$$= 0 \quad (14)$$

Since $\operatorname{sgn}[\cos(\omega_r t)]$ is 1 from 0 to $T/4$ and -1 from $T/4$ to $T/2$, thus yielding even contributions over the interval with different signs cancelling each other out.

Let's determine a_n . We split the integral up using $\operatorname{sgn}[\cos(\omega_r t)]$:

$$a_n = \frac{4}{T} \left(\left(\int_0^{T/4} dt \cos(2\pi n t/T) \right) - \left(\int_{T/4}^{T/2} dt \cos(2\pi n t/T) \right) \right) \quad (15)$$

$$= \frac{4T}{2\pi n T} \left([\sin(2\pi n t/T)]_0^{T/4} - [\sin(2\pi n t/T)]_{T/4}^{T/2} \right) \quad (16)$$

$$= \frac{2}{\pi n} (2 \sin(\pi n/2) - \sin(\pi n)) \quad (17)$$

For $n \in \mathbb{N}$ $\sin(\pi n) = 0 \forall n$. Thus:

$$a_n = \frac{4}{\pi n} \sin(\pi n/2) \quad (18)$$

For even harmonics $n = 2k$, $k \in \mathbb{N}$, we have:

$$a_{n \rightarrow 2k} = \frac{4}{\pi 2k} \sin(\pi 2k/2) = \frac{2}{\pi k} \sin(\pi k) = 0 \quad (19)$$

For odd harmonics, $n = 2k + 1$, we have:

$$a_{n \rightarrow 2k+1} = \frac{4}{\pi(2k+1)} \sin(\pi(2k+1)/2) \quad (20)$$

$$= \frac{4}{\pi(2k+1)} \sin(\pi k + \pi/2) \quad (21)$$

$$= \frac{4}{\pi(2k+1)} \cos(\pi k) \quad (22)$$

$$= \frac{4}{\pi(2k+1)} (-1)^k \quad (23)$$

Inserting a_n and doing a change of indices, $V_r(t)$ is given as:

$$V_r(t) = A_r \left(0 + \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)} \cos((2k+1)\omega_r t) \right) \quad (24)$$

$$= \frac{4A_r}{\pi} (\cos(\omega_r t) - \cos(3\omega_r t)/3 + \dots) \quad (25)$$

Assembling the pieces

Using our new expression for $V_r(t)$, we get:

$$X(t) = V_s(t)V_r(t) \quad (26)$$

$$= \frac{4A_s A_r}{\pi} \cos(\omega_s t + \phi) \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)} \cos((2k+1)\omega_r t) \quad (27)$$

$$= \frac{4A_s A_r}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)} \cos(\omega_s t + \phi) \cos((2k+1)\omega_r t) \quad (28)$$

$$(29)$$

Using $\cos(x)\cos(y) = \frac{1}{2}(\cos(x-y) + \cos(x+y))$, we get:

$$X(t) = \frac{2A_s A_r}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)} [\cos((\omega_s - (2k+1)\omega_r)t + \phi) + \cos((\omega_s + (2k+1)\omega_r)t + \phi)] \quad (30)$$

Let $\delta = \omega_s - \omega_r$ and $\Gamma = A_s A_r$. If the low-pass filter bandwidth is narrow enough, then only the term nearest the fundamental-frequency detuning, δ , survives, and we have:

$$X(t) \approx \frac{2\Gamma}{\pi} \cos(\delta t + \phi) \quad (31)$$

Setting $\delta = 0$, we obtain:

$$X(t)_{\delta=0} \approx \frac{2\Gamma}{\pi} \cos(\phi) \quad (32)$$

Bode plot of stages 1–3

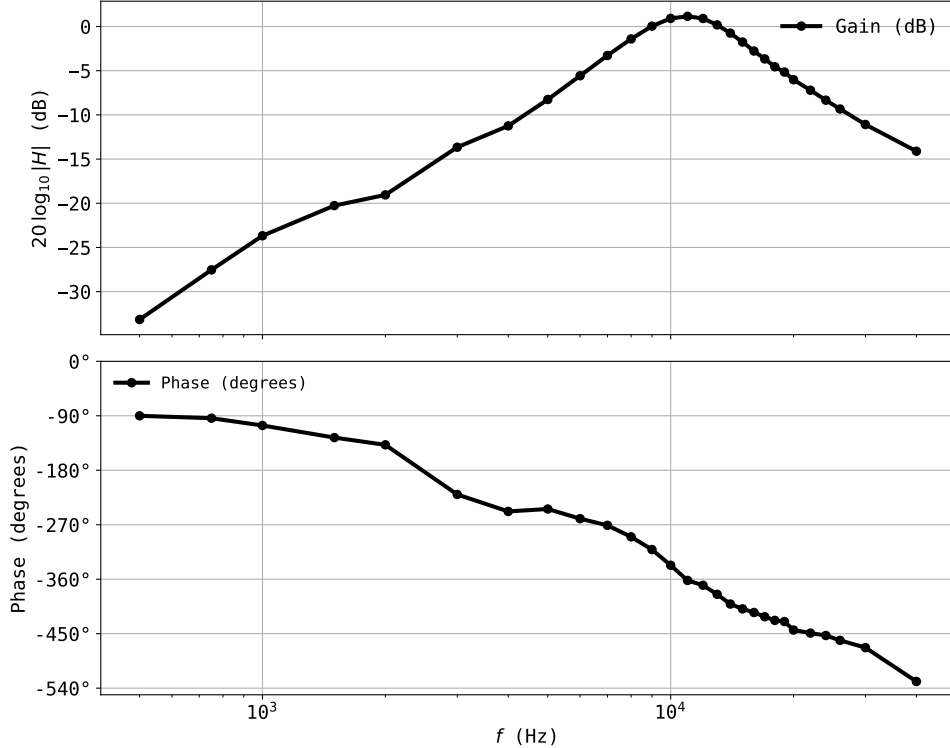


FIG. I. (Top) Bode magnitude plot of measurements from experiment 2 (same as Fig. 2(c)). (Bottom) Bode phase plot of the measurements from experiment 2.

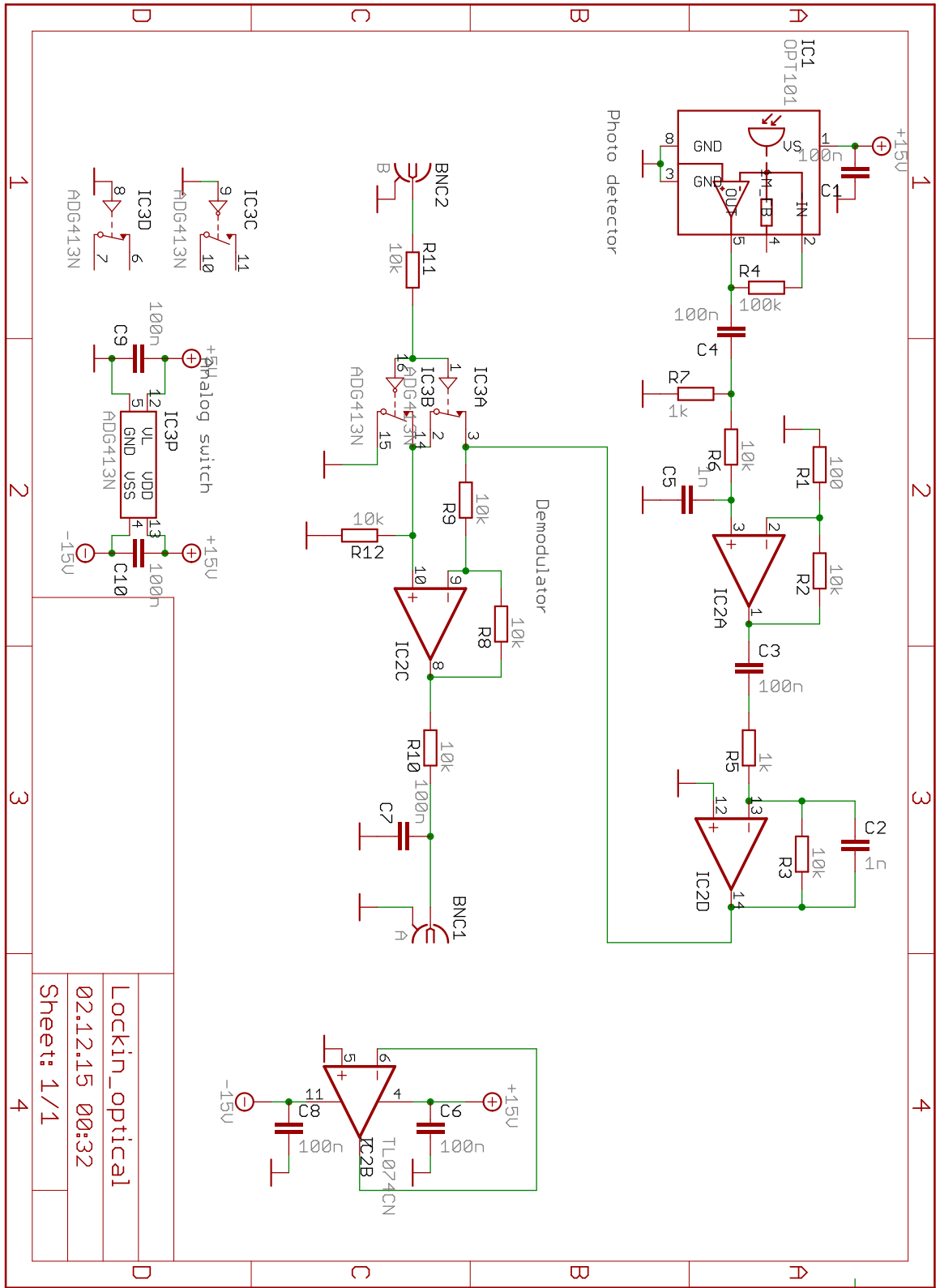


FIG. II. Detailed lumped-element schematic of the constructed lock-in amplifier.

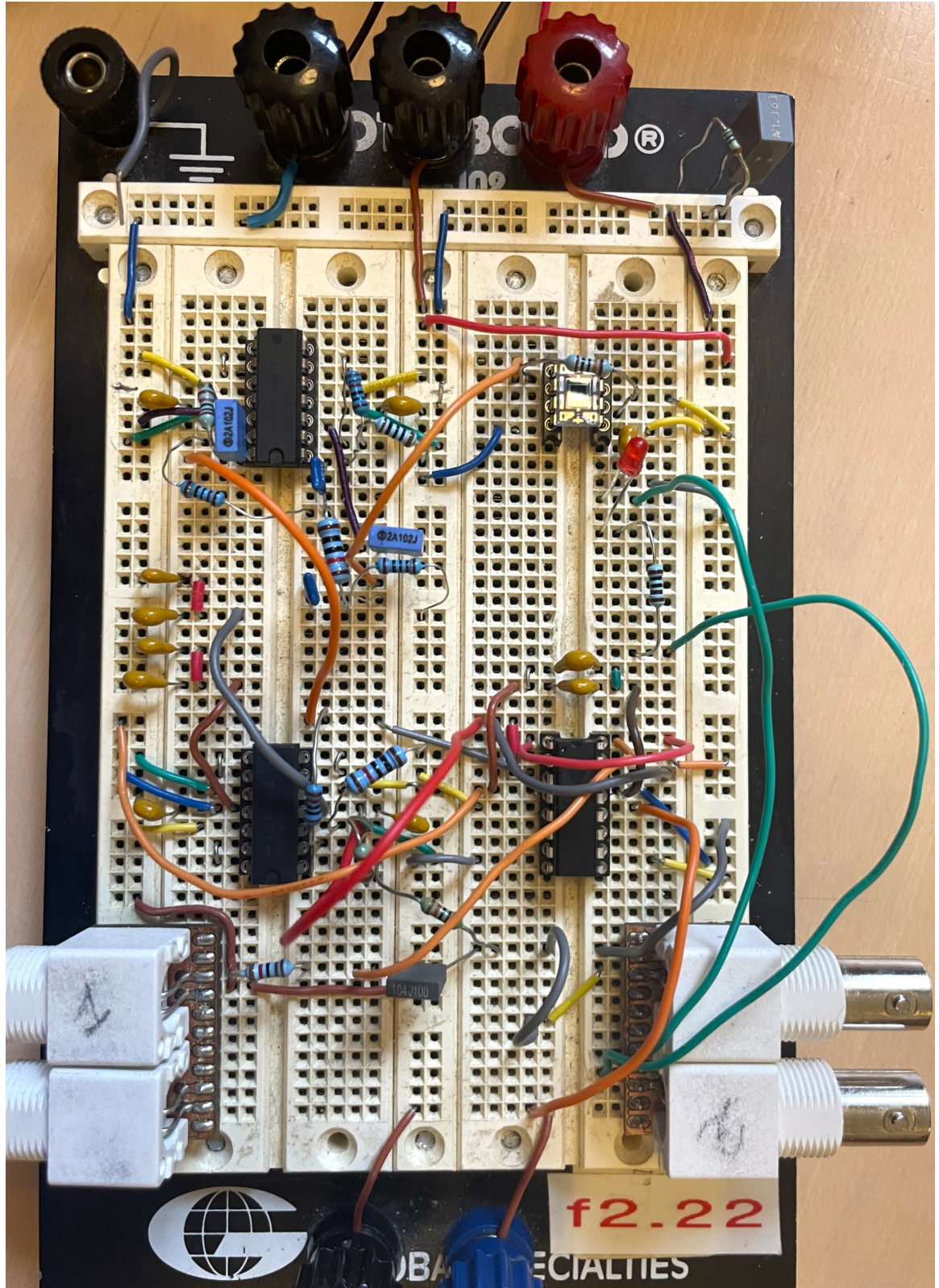


FIG. III. Breadboard implementation of the lock-in amplifier. The breadboard contains several passive and active components; the utilized ICs are identified here as follows: **Top left:** TL074 op-amp IC used in stages 2 and 3 (IC2A and IC2D in Fig. II). **Bottom left:** TL074 op-amp IC used in stage 4 (IC2C in Fig. II). **Top right:** OPT101 photodiode providing the signal voltage $V_s(t)$. **Bottom right:** ADG413N analog switch IC used in stage 4 (IC3A and IC3B in Fig. II). Additional views of the setup are available in the project GitHub repository.